

Deterministic Interpretation of Two Symbolic Verifications

The Vacuum Mirror (Thm. 6.23) and the Radiative Quartics (Prop. 7.6)

Electromagnetic Sphere monograph — machine-checked supplement

Provenance and reading rules. Every physical statement in this report is anchored to a *specific symbolic invariant* produced by direct execution of the two scripts `verify_thm_6_23_vacuum_mirror.py` and `verify_prop_7_6_radiative_quartics.py`. The execution engine is SymPy (exact symbolic algebra over the explicit 8×8 real representation of $\text{Cl}(3,3)$) with a NumPy Hermitian eigensolver for the numerical spectrum. *No claim is made that is not pinned to an output line*; where an output is identically zero, no non-zero physical effect is inferred from it. Status flags follow the monograph: [EXACT] (algebraic identity, exact), [DERIVED] (consequence proved from exact identities), [OPEN] (mechanism shown, full proposition not closed). Both runs terminated with all checks passing: **Thm 6.23 block 23/23, Prop 7.6 block 14/14**, exit code 0.

0 Shared object: the 8×8 real representation of $\text{Cl}(3,3)$

Both verifications live in one fixed representation, so the anchors are commensurable. On $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$ with split signature $(3,3)$ the six generators are built from the real 2×2 seeds

$$s_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad s_1^2 = s_3^2 = +\mathbf{1}_2, \quad \epsilon^2 = -\mathbf{1}_2,$$

via Kronecker products, giving three *temporal* generators ($e_i^2 = +\mathbf{1}$) and three *spatial* ones ($e_i^2 = -\mathbf{1}$):

$$\begin{aligned} e_1 &= s_1 \otimes \mathbf{1} \otimes \mathbf{1}, & e_2 &= s_3 \otimes s_1 \otimes \mathbf{1}, & e_3 &= s_3 \otimes s_3 \otimes s_1, & (e_i^2 &= +\mathbf{1}_8) \\ e_4 &= \epsilon \otimes \mathbf{1} \otimes \mathbf{1}, & e_5 &= s_3 \otimes \epsilon \otimes \mathbf{1}, & e_6 &= s_3 \otimes s_3 \otimes \epsilon, & (e_i^2 &= -\mathbf{1}_8) \end{aligned}$$

with metric $\eta = \text{diag}(+, +, +, -, -, -)$. From these the script derives, once and for all,

$$\begin{aligned} \widehat{W} &= \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3) && \text{kernel clock,} \\ \Gamma_a &= \widehat{W} e_{3+a} \quad (a = 1, 2, 3) && \text{interior Dirac matrices,} \\ \mathbf{V} &= e_4 e_5 e_6 && \text{covariant-mass trivector,} \\ I &= e_1 e_2 e_3 e_4 e_5 e_6 && \text{pseudoscalar,} \\ \omega^k &= e_k e_{k+3} \quad (k = 1, 2, 3) && \text{boost-Cartan / octant generators,} \\ \Phi &= \sum_k \phi_k \omega^k && \text{bivector condensate.} \end{aligned}$$

Trace convention. The literal matrix trace with $\text{Tr } \mathbf{1} = 8$ is used everywhere, and is *pinned*, *not assumed*: the run re-checks the monograph Killing norms

$$\text{Tr}[(e_1 e_2)^2] = -8 \quad (\text{compact}), \quad \text{Tr}[(e_1 e_4)^2] = +8 \quad (\text{boost}). \quad (\text{both OK})$$

The normalized trace $\text{tr} = \frac{1}{8} \text{Tr}$ ($\text{tr } \mathbf{1} = 1$) is referenced only as a convention bridge.

Part A Theorem 6.23 — the vacuum is the chiral mirror of matter

A.1 Raw execution-data anchors (literal outputs)

Each row is a symbolic equality returned identically true ($= Z_8$ means the residual matrix is the 8×8 zero matrix) or a numeric eigenvalue set. These are the *only* facts subsequent sections are permitted to use.

#	Symbolic anchor (literal output)	Object / meaning	Flag
A1	$\{e_M, e_N\} = 2 \eta_{MN} \mathbf{1}_8$	$\text{Cl}(3, 3)$ relations, signature $(3, 3)$	[EXACT]
A2	$\widehat{W}^2 = \mathbf{1}_8$	clock is an involution	[EXACT]
A3	$\Gamma_a^\top = \Gamma_a, \quad \Gamma_a^2 = +\mathbf{1}_8$	Dirac matrices symmetric, unit square	[EXACT]
A4	$\{\Gamma_i, \Gamma_j\} = 0 \ (i \neq j)$	spatial Clifford triple	[EXACT]
A5	$\{\mathbf{V}, \Gamma_a\} = 0$	trivector <i>anticommutes</i> with all kinetic terms	[EXACT]
A6	$I^2 = +\mathbf{1}_8$	pseudoscalar is an involution	[EXACT]
A7	$[I, \Gamma_a] = 0$	I <i>commutes</i> with kinetic terms	[EXACT]
A8	$\{I, \mathbf{V}\} = 0$	I <i>anticommutes</i> with the mass term	[EXACT]
A9	$\text{Tr } H(m) = 0$	single-particle Hamiltonian traceless	[EXACT]
A10	$H(m)^2 = (p_1^2 + p_2^2 + p_3^2 + m^2) \mathbf{1}_8$	squared Hamiltonian is a scalar	[EXACT]
A11	$I H(m) I = H(-m)$	mirror identity	[EXACT]
A12	$\text{spec } H = \{\pm\sqrt{20.25}\} = \{\pm 4.5\}, 4+4$	exact-rational test $p = (2, -1, 3), m = \frac{5}{2}$	[DERIVED]
A13	$\text{spec } H(m) = \text{spec } H(-m)$	isospectral $\pm m$ branches	[DERIVED]
A14	$\omega^2 = p^2 + m^2 = 20.25 > 0$	real frequencies, no imaginary mode	[DERIVED]
A15	$[\omega_i, \omega_j] = 0$	octant generators mutually commute	[EXACT]
A16	$\{\mathbf{V}, \omega^k\} = 0$	$\mathbf{V}$ flips every octant to its antipode	[EXACT]

Here $H(m) = p_1 \Gamma_1 + p_2 \Gamma_2 + p_3 \Gamma_3 + m \mathbf{V}$ with real p_i, m .

A.2 Mathematical symmetries (what the algebra allows)

A grading by $\mathbf{V}$ and by I . A5 and A8 state two independent $\mathbb{Z}_2$ gradings of the operator algebra. Under conjugation by the trivector, the kinetic part is odd and—via A10—the square is even; under conjugation by the pseudoscalar (A7–A8) the kinetic part is *even* and the mass part is *odd*. The kinetic operators and the mass operator therefore sit in orthogonal eigenspaces of the same two involutions; this is the precise algebraic content of “chirality”.

The square closes on a scalar (A10). Because A3, A4, A5 give $\Gamma_a^2 = \mathbf{1}$, $\{\Gamma_i, \Gamma_j\} = 0$ and $\{\mathbf{V}, \Gamma_a\} = 0$, every cross term in H^2 cancels and only the squares survive:

$$H(m)^2 = \sum_a p_a^2 \Gamma_a^2 + m^2 \mathbf{V}^2 = (p_1^2 + p_2^2 + p_3^2 + m^2) \mathbf{1}_8.$$

Mathematically this is allowed *only* because the mass generator anticommutes with the kinetic generators (A5): a commuting mass term would leave a non-scalar cross term.

The mirror is exact, not approximate (A11). Since I commutes with each Γ_a (A7) and anticommutes with $\mathbf{V}$ (A8) while $I^2 = \mathbf{1}$ (A6),

$$I H(m) I = \sum_a p_a I \Gamma_a I + m I \mathbf{V} I = \sum_a p_a \Gamma_a - m \mathbf{V} = H(-m).$$

The involution I implements the discrete map $m \mapsto -m$ *exactly*, with no residual term, so the two sign branches are unitarily equivalent.

Spectral consequences (A12–A14). From A9–A10 the characteristic data are fixed: tracelessness forces a symmetric spectrum about 0, and $H^2 = \omega^2 \mathbf{1}$ forces every eigenvalue to be $\pm\omega$ with $\omega = \sqrt{p^2 + m^2}$, each with multiplicity 4 in the 8-dimensional space. A13 is then automatic from A11 (conjugation preserves spectrum). $\omega^2 > 0$ in A14 is an algebraic identity, not a fit.

Antipodal action on octants (A15–A16). The three ω^k commute (A15), so they are simultaneously diagonalizable and label the eight sign-sectors (“octants”) by $(\pm, \pm, \pm)$. Equation A16, $\{V, \omega^k\} = 0$, says V reverses every label at once: it is the antipodal map on the octant lattice.

A.3 Physical interpretation & boundary limits

What the anchors license.

- **Covariant (not merely additive) mass.** Anchor A5 ($\{V, \Gamma_a\} = 0$) is the statement that V enters H exactly as a Dirac mass: it gaps the spectrum through the m^2 term in A10 rather than shifting it linearly. The mass is carried by the *grade-3 volume trivector*, distinct from the gauge mechanism of Part B.
- **The vacuum branch has negative mass and is stable.** A11 identifies the $m \mapsto -m$ partner $H(-m)$ as the image of $H(m)$ under the chiral involution I . A13 shows the two branches are isospectral and A14 shows $\omega^2 = p^2 + m^2 > 0$ on both. *Anchored conclusion:* the negative-mass (vacuum) sector is the exact CPT-type mirror of the positive-mass (matter) sector — kinetic and mass signs flip *together* under one involution — and it carries real frequencies. There is **no tachyon**: a tachyonic branch would require $\omega^2 < 0$, which A14 forbids (this is precisely the “output is positive $\Rightarrow$ no imaginary mode” direction of the reading rule).
- **Octant antipodal pairing.** A15–A16 give the geometric picture: matter and vacuum occupy antipodal octants of M^6 , exchanged by the single map V .

Where the result stops (boundary limits).

- **Single-particle, free level only.** Every anchor is an algebraic property of the one-body operator H or of the generators. *No interaction, loop, or many-body statement is anchored here*; stability is the statement $\omega^2 \geq 0$ of the free dispersion, not a proof of vacuum stability under dynamics.
- **Hamiltonian frame, full Lorentz covariance open.** A5–A11 are stated for the 3+1 reduced Hamiltonian built from $\Gamma_a = \widehat{W}e_{3+a}$. The covariance of the mass term under the full split-signature symmetry is *not* settled by these outputs and remains **[OPEN]** in the monograph.
- **Magnitudes absent.** m is a free symbol; no anchor fixes its value. The result is structural (“the spectrum is $\pm\sqrt{p^2 + m^2}$, 4+4”), with the scale of m outside scope.

Part B Proposition 7.6 — radiative quartics and the gauge mass

Notation for the invariants of the condensate $\Phi = \sum_k \phi_k \omega^k$:

$$P_4 = \sum_i \phi_i^4, \quad P_{2,2} = \sum_{i < j} \phi_i^2 \phi_j^2, \quad (\varepsilon \cdot \phi) = \sum_i \varepsilon_i \phi_i, \quad \varepsilon_i \in \{+1, -1\}.$$

B.1 Raw execution-data anchors (literal outputs)

#	Symbolic anchor (literal output)	Meaning	Flag
B0	$\text{Tr}[(e_1 e_2)^2] = -8, \quad \text{Tr}[(e_1 e_4)^2] = +8$	trace convention pinned	[EXACT]
B1	$\text{Tr} \Phi^2 = \sum_{\epsilon} (\epsilon \cdot \phi)^2 = 8 \sum_i \phi_i^2$	quadratic octant sum rule	[EXACT]
B2	$\text{Tr} \Phi^4 = \sum_{\epsilon} (\epsilon \cdot \phi)^4 = 8 P_4 + 48 P_{2,2}$	quartic octant sum rule	[EXACT]
B3	$(\text{Tr} \Phi^2)^2 = 64 P_4 + 128 P_{2,2}$	competing quartic invariant	[EXACT]
B4	$\det \begin{pmatrix} 8 & 48 \\ 64 & 128 \end{pmatrix} = -2048 \neq 0$	$\text{Tr} \Phi^4, (\text{Tr} \Phi^2)^2$ independent	[EXACT]
B5	$(\epsilon \cdot \phi)^4 \big _{\phi_1 \rightarrow -\phi_1} = (\epsilon \cdot \phi)^4$	octant quartic parity-even	[EXACT]
B6	$\phi_1 \phi_2 \phi_3 \big _{\phi_1 \rightarrow -\phi_1} = -\phi_1 \phi_2 \phi_3$	chiral cubic parity-odd	[EXACT]
B7	$\text{Tr}(\Phi^3 I) = -48 \phi_1 \phi_2 \phi_3$	cubic matrix trace (nonzero)	[EXACT]
B8	$\text{spec ad}_{\Phi} = \{0, 0, 0\} \cup \{2(\pm \phi_i \pm \phi_j)\}$	gauge masses = $\mathfrak{so}(3, 3)$ roots	[DERIVED]
B9	$\text{Tr}(\text{ad}_{\Phi}^4) = 128 P_4 + 384 P_{2,2}$	gauge $\sum_a m_a^4$	[DERIVED]
B10	$\text{Tr}(\text{ad}_{\Phi}^4) = 4 \text{Tr} \Phi^4 + \frac{3}{2} (\text{Tr} \Phi^2)^2$	$\propto 8 \text{Tr} \Phi^4 + 3 (\text{Tr} \Phi^2)^2$	[DERIVED]

At the numerical test point $\phi = (1, 2, 4)$ the literal ad_{Φ} eigenvalue list is

$$\{-12, -10, -6, -6, -4, -2, 0, 0, 0, 2, 4, 6, 6, 10, 12\},$$

which matches the $\mathfrak{so}(3, 3)$ root prediction $2(\pm \phi_i \pm \phi_j)$ plus three Cartan zeros term-by-term (anchor B8, **OK**).

B.2 Mathematical symmetries (what the algebra allows)

Two truly distinct quartic invariants (B2–B4). The fermionic octant loop returns the single invariant $\text{Tr} \Phi^4 = 8P_4 + 48P_{2,2}$ (anchor B2). The competing invariant $(\text{Tr} \Phi^2)^2 = 64P_4 + 128P_{2,2}$ (B3) is a *different* vector in the two-dimensional space spanned by $\{P_4, P_{2,2}\}$: the change-of-basis determinant is $-2048 \neq 0$ (B4). Therefore $\text{Tr} \Phi^4$ has *zero component* along $(\text{Tr} \Phi^2)^2$ when the two are taken as the basis pair — the algebra forbids one from being a multiple of the other.

Parity is the operative selection rule (B5–B7). The one-loop (Coleman–Weinberg) potential of the fermion octant sector is a function of the squared masses, i.e. of the octant power sums $\sum_{\epsilon} (\epsilon \cdot \phi)^n$, which are *even* under every $\phi_i \rightarrow -\phi_i$ (B5). The chiral cubic invariant $\phi_1 \phi_2 \phi_3$ is *odd* (B6). An even function cannot generate an odd monomial, so the coefficient of $\phi_1 \phi_2 \phi_3$ in the radiative potential is forced to zero *by symmetry*.

Reading-rule caveat. The cubic matrix trace $\text{Tr}(\Phi^3 I) = -48 \phi_1 \phi_2 \phi_3$ (B7) is **not** zero. The vanishing of the radiative cubic is therefore anchored to the parity pair (B5) $\wedge$ (B6), *not* to “ $\text{Tr}(\Phi^3 I) = 0$ ”. The cold inference is: *even loop integrand $\times$ odd invariant $\Rightarrow$ zero coefficient.*

The gauge sector is in the adjoint (B8). The gauge bosons are bivectors, so the condensate masses them through ad_{Φ} , not through left multiplication. Anchor B8 states that the spectrum of ad_{Φ} on the 15 bivectors of $\mathfrak{so}(3, 3)$ is exactly the root system: three Cartan zeros (the directions commuting with Φ , anchor A15 re-used) and the twelve roots $2(\pm \phi_i \pm \phi_j)$. This is verified eigenvalue-by-eigenvalue at $\phi = (1, 2, 4)$.

The gauge loop supplies the second invariant (B9–B10). Summing the fourth powers of the root masses gives $\text{Tr}(\text{ad}_{\Phi}^4) = 128P_4 + 384P_{2,2}$ (B9), which decomposes as

$$\text{Tr}(\text{ad}_{\Phi}^4) = 4 \text{Tr} \Phi^4 + \frac{3}{2} (\text{Tr} \Phi^2)^2 \iff \propto 8 \text{Tr} \Phi^4 + 3 (\text{Tr} \Phi^2)^2,$$

i.e. ratio $(\text{Tr} \Phi^2)^2 : \text{Tr} \Phi^4 = 3 : 8$ (anchor B10). Unlike the fermion result, this has a *non-zero* $(\text{Tr} \Phi^2)^2$ component.

B.3 Physical interpretation & boundary limits

What the anchors license.

- **Two independent quartic couplings B, C are required, and they have distinct origins.** Writing the effective quartic potential as $B \operatorname{Tr} \Phi^4 + C (\operatorname{Tr} \Phi^2)^2$, anchors B2 and B4 show the fermion octant loop feeds only the B -type invariant $(\operatorname{Tr} \Phi^4)$ and contributes *zero* to C — this is the genuine “output has no $(\operatorname{Tr} \Phi^2)^2$ component” reading, certified by the nonzero determinant. Anchor B10 shows the gauge (adjoint) loop is what supplies the C coupling, in the fixed ratio 3 : 8.
- **No radiative chiral cubic.** Anchors B5–B6 forbid a $\phi_1 \phi_2 \phi_3$ term in the one-loop potential on parity grounds. (Proposition 7.7.)
- **Gauge masses are organized by the $\mathfrak{so}(3, 3)$ root lattice.** Anchor B8 gives the boson mass² spectrum as the squared roots $[2(\pm\phi_i \pm \phi_j)]^2$, with three exactly massless directions (the Cartan) — the unbroken rank-3 subgroup.

Where the result stops (boundary limits).

- **Proposition 7.7 retains [OPEN] status.** The parity anchors close the *real Coleman–Weinberg* channel only. They do not exclude a cubic from non-perturbative effects, explicit tree-level terms, or anomaly-type contributions; the monograph keeps the proposition open.
- **Structure, not magnitude.** The anchors fix the *ratio* 3 : 8 and the *form* $B \operatorname{Tr} \Phi^4 + C (\operatorname{Tr} \Phi^2)^2$. The overall scale — the $\mathcal{O}(g^4/64\pi^2)$ prefactor and the couplings g, v, g_3 — is an input, not an output. “Structure complete, magnitudes open.”
- **Convention sensitivity is explicit.** The numbers (−48; ratio 3 : 8) are tied to the matrix trace $\operatorname{Tr} \mathbf{1} = 8$, pinned by B0. In the normalized trace $\operatorname{tr} = \frac{1}{8} \operatorname{Tr}$ the same physics reads $-6 \phi_1 \phi_2 \phi_3$ and ratio 3 : 1; the ratio shifts because $\operatorname{Tr} \Phi^4$ is linear in the trace while $(\operatorname{Tr} \Phi^2)^2$ is quadratic. Quoting one set of numbers under the other symbol (an error in an earlier draft) is the only inconsistency the run was built to catch, and it is resolved in favour of the $\operatorname{Tr} \mathbf{1} = 8$ values.
- **No colour sector.** The construction is internal to $\mathfrak{so}(3, 3)$; the $SU(3)_c$ colour group is not present in these outputs and is outside scope.

Summary ledger

Statement	Anchor(s)	Flag	One-line reading
Cl(3, 3) representation valid	A1, B0	[EXACT]	generators and trace norms correct
Covariant fermion mass (trivector)	A5, A10	[EXACT]	$\{\mathbf{V}, \Gamma_a\} = 0$ gaps the spectrum
Balanced spectrum $\pm\sqrt{p^2 + m^2}$	A9, A10	[EXACT]	traceless, scalar square
Vacuum = chiral mirror of matter	A11	[EXACT]	$IH(m)I = H(-m)$, exact involution
Negative-mass branch is stable	A13, A14	[DERIVED]	isospectral, $\omega^2 > 0$, no tachyon
Octant antipodal pairing	A15, A16	[EXACT]	$\mathbf{V}$ reverses all octant signs
Octant quartic sum rule	B2	[EXACT]	$\operatorname{Tr} \Phi^4 = 8P_4 + 48P_{2,2}$
Fermion loop $\rightarrow B$ only, $C = 0$	B2, B4	[EXACT]	quartics independent, $\det = -2048$
No radiative chiral cubic	B5, B6	[EXACT]	even loop $\times$ odd invariant = 0
Cubic remains a proposition	B5–B7	[OPEN]	parity closes CW channel only
Gauge mass = $\mathfrak{so}(3, 3)$ roots	B8	[DERIVED]	$\{0^3\} \cup \{2(\pm\phi_i \pm \phi_j)\}$
Gauge loop supplies C , ratio 3:8	B9, B10	[DERIVED]	$4 \operatorname{Tr} \Phi^4 + \frac{3}{2} (\operatorname{Tr} \Phi^2)^2$

Net. The two runs jointly establish, at the algebraic and free-spectral level, that (i) matter and vacuum are an exact chiral-mirror pair with a stable negative-mass branch (Thm 6.23), and (ii) the scalar potential needs two quartic couplings whose $(\text{Tr } \Phi^2)^2$ part is generated by the gauge adjoint loop in the ratio 3 : 8, with the chiral cubic forbidden by parity (Prop 7.6/7.7). All magnitude, full-covariance, ghost, and colour questions remain outside the scope of these outputs.